# Cabinet schematic — Pump cabinet

## Pump cabinet

Two direct-on-line motor starters, controlled and monitored by a WAGO 750 PLC through a relay interface board.

| Revision | Date | Description | Created | Checked | Approved |
|---|---|---|---|---|---|
| 1.1 | 2026-09-01 | First release | SK | | |
| 1.2 | 2026-09-23 | Overload feedback to the PLC | SK | | |
| 1.3 | 2026-10-02 | TeSys D parts, 24V and GND rails, IEC wire colours | SK | | |
| 1.4 | 2026-10-05 | Unit tags, links, busbars and rail bonds, energy on fed supplies | SK | | |
| 1.5 | 2026-10-06 | Three phases on four terminals each, pump n on terminal n+2 | SK | | |
| 1.6 | 2026-10-06 | Owner ruling 2026-10-06, a cabinet is a unit: the place C1 is gone. The change list reads No changes. because the listing drops a place that holds the whole unit, baseline L2, kept by design (REL-DIFF-PLACE) | SK | | |

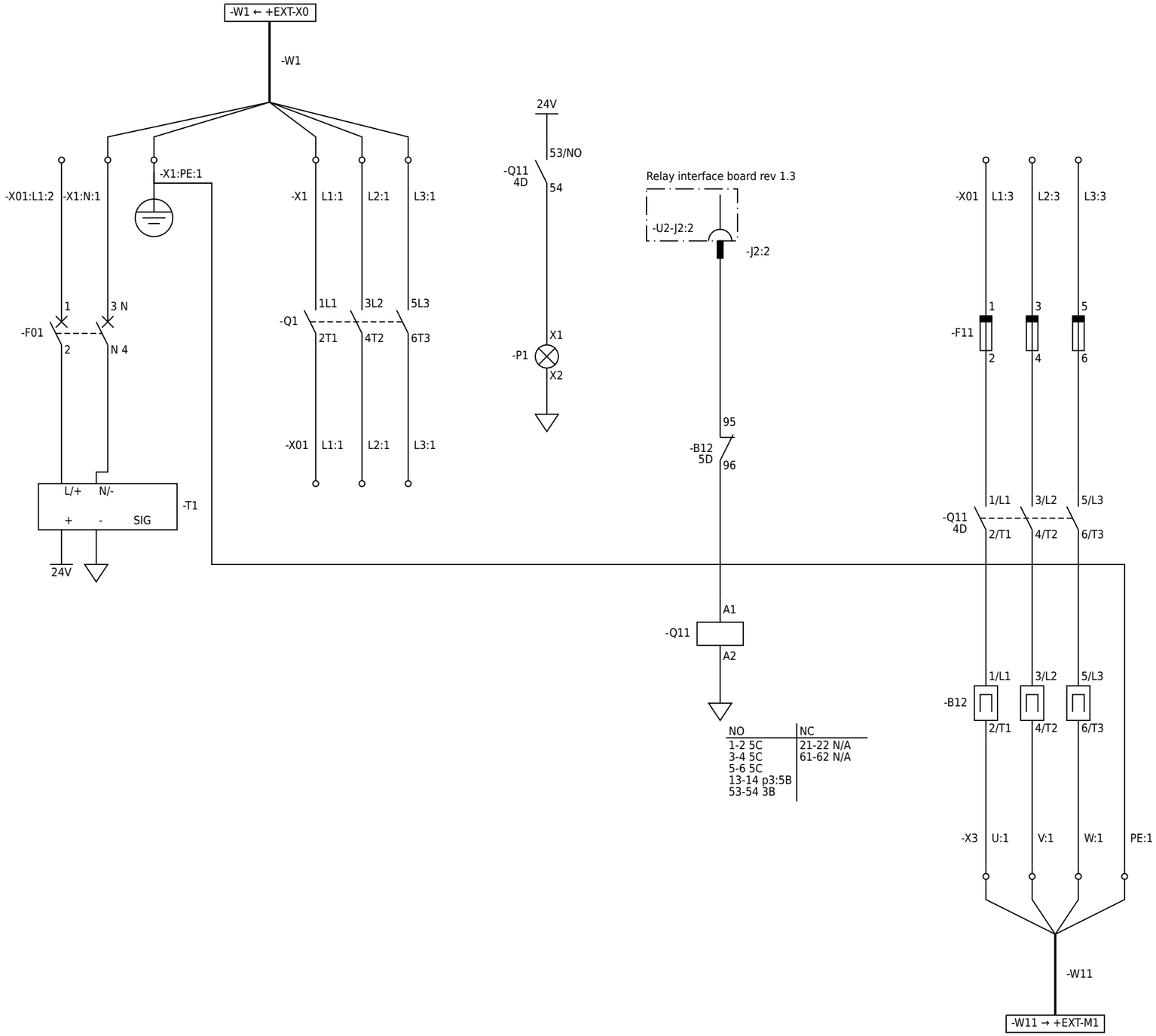
| | | | | | | |
|---|---|---|---|---|---|---|
| Title<br>**Pump cabinet** | | | Number<br>SKX-PC-2 | Revision<br>1.6 | Revision date<br>2026-10-06 | |
| Customer | Author<br>fransys-examples | Page title<br>Cover | Scope<br>Pump cabinet | Sheet | Page<br>1 of 8 | |

# Notes

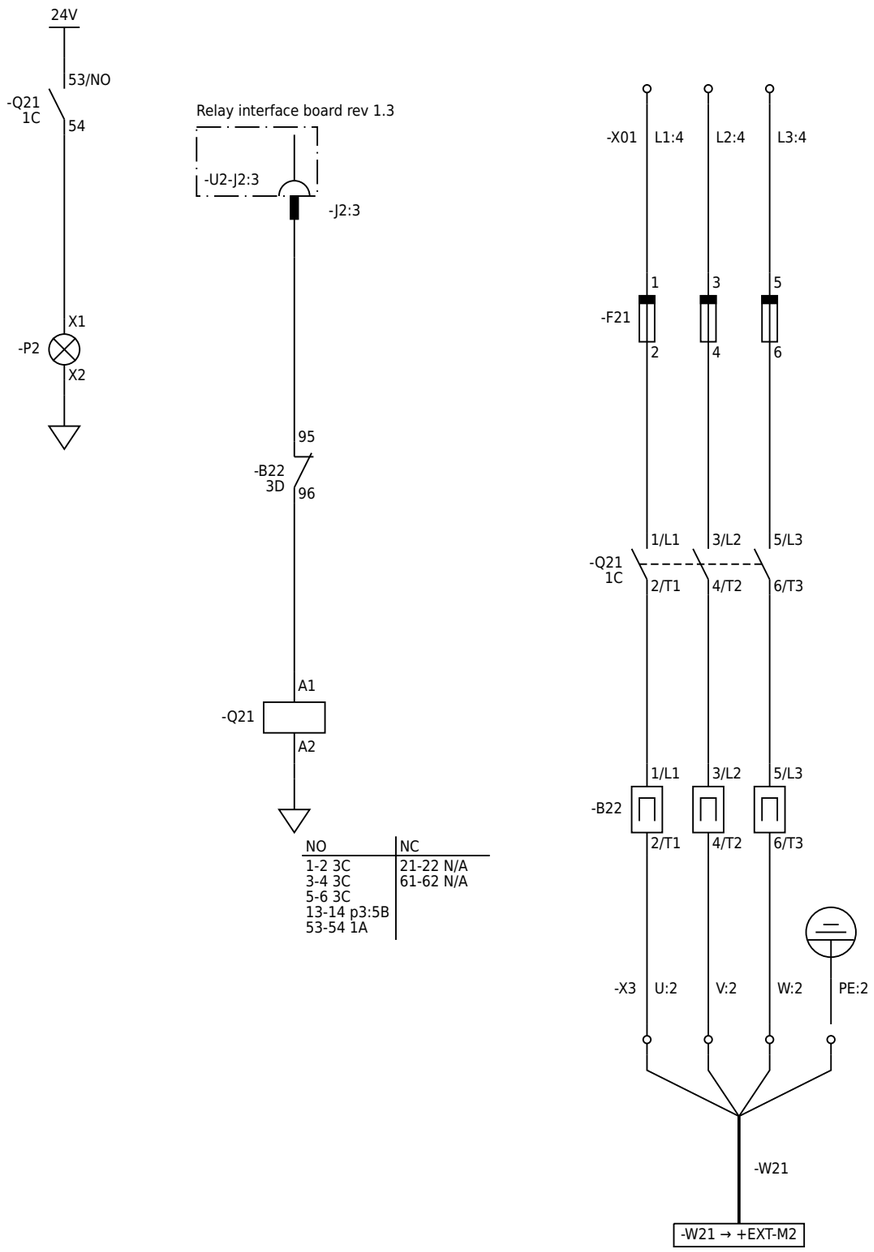
The upstream board protects the incoming cable with a C16 breaker. The cable lands there on terminals +EXT-X0, the only part of the upstream board in this model.

The relay interface board -U2 has its own drawing set. These pages show it only as its two headers inside a dash-dot outline.

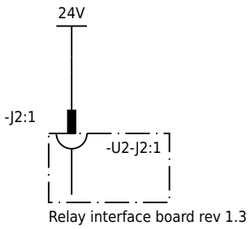
| | | | | | | |
|---|---|---|---|---|---|---|
| Title<br>**Pump cabinet** | | | Number<br>SKX-PC-2 | Revision<br>1.6 | Revision date<br>2026-10-06 | |
| Customer | Author<br>fransys-examples | Page title<br>Notes | Scope<br>Pump cabinet | Sheet | Page<br>2 of 8 | |



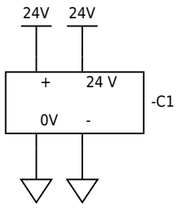

| | | | | | |
|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.6 | Revision date: 2026-10-06 |
| Customer | Author: fransys-examples | Page title: Control supply, Incoming supply, Pump 1 | Scope: Pump cabinet | Sheet: 1 / 3 | Page: 3 of 8 |

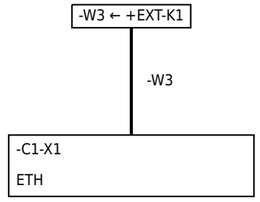


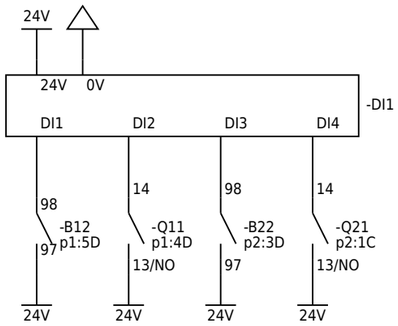

| | | | | | |
|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.6 | Revision date: 2026-10-06 |
| Customer | Author: fransys-examples | Page title: Pump 2 | Scope: Pump cabinet | Sheet: 2 / 3 | Page: 4 of 8 |

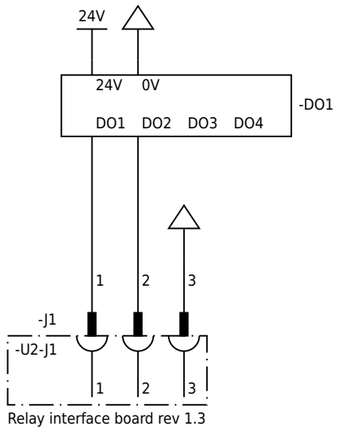



Relay interface board rev 1.3




Relay interface board rev 1.3

| | | | | | | |
|---|---|---|---|---|---|---|
| Title<br>**Pump cabinet** | | | Number<br>SKX-PC-2 | Revision<br>1.6 | Revision date<br>2026-10-06 | |
| Customer | Author<br>fransys-examples | Page title<br>WAGO PLC | Scope<br>Pump cabinet | Sheet<br>3 / 3 | Page<br>5 of 8 | |

## -DI1

| Channel designation | Signal | Wired to | Signal name |
|---|---|---|---|
| -DI1:1 | di | -B12:98 | P1_OVERLOAD |
| -DI1:2 | di | -Q11:14 | P1_RUNNING |
| -DI1:3 | di | -B22:98 | P2_OVERLOAD |
| -DI1:4 | di | -Q21:14 | P2_RUNNING |

## -DO1

| Channel designation | Signal | Wired to | Signal name |
|---|---|---|---|
| -DO1:1 | do | -J1:1 | P1_RUN |
| -DO1:2 | do | -J1:2 | P2_RUN |
| -DO1:3 | do | | |
| -DO1:4 | do | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.6 | Revision date: 2026-10-06 | |
| Customer | Author: fransys-examples | Page title: PLC list | Scope: Pump cabinet | Sheet | Page: 6 of 8 | |

## -X01

| Designation | Group | Index | Side A | Bridge |
|---|---|---|---|---|
| -X01:L1:1 | L1 | 1 | -Q1:2T1 | |
| -X01:L1:2 | L1 | 2 | -F01:1 | |
| -X01:L1:3 | L1 | 3 | -F11:1 | |
| -X01:L1:4 | L1 | 4 | -F21:1 | |
| -X01:L2:1 | L2 | 1 | -Q1:4T2 | |
| -X01:L2:2 | L2 | 2 | | |
| -X01:L2:3 | L2 | 3 | -F11:3 | |
| -X01:L2:4 | L2 | 4 | -F21:3 | |
| -X01:L3:1 | L3 | 1 | -Q1:6T3 | |
| -X01:L3:2 | L3 | 2 | | |
| -X01:L3:3 | L3 | 3 | -F11:5 | |
| -X01:L3:4 | L3 | 4 | -F21:5 | |

## -X1

| Designation | Group | Index | Side A | Side B |
|---|---|---|---|---|
| -X1:L1:1 | L1 | 1 | -Q1:1L1 | +EXT-X0:L1:1 |
| -X1:L2:1 | L2 | 1 | -Q1:3L2 | +EXT-X0:L2:1 |
| -X1:L3:1 | L3 | 1 | -Q1:5L3 | +EXT-X0:L3:1 |
| -X1:N:1 | N | 1 | -F01:3 N | +EXT-X0:N:1 |
| -X1:PE:1 | PE | 1 | -X3:PE:1; -X3:PE:2 | +EXT-X0:PE:1 |

## -X2

| Designation | Group | Index | Side A | Bridge |
|---|---|---|---|---|
| -X2:24V:1 | 24V | 1 | -T1:+ | |
| -X2:24V:2 | 24V | 2 | -C1:24 V | |
| -X2:24V:3 | 24V | 3 | -C1:+ | |
| -X2:24V:4 | 24V | 4 | -J2:1 | |
| -X2:24V:5 | 24V | 5 | -Q11:53/NO | |
| -X2:24V:6 | 24V | 6 | -B12:97 | |
| -X2:24V:7 | 24V | 7 | -Q11:13/NO | |
| -X2:24V:8 | 24V | 8 | -Q21:53/NO | |
| -X2:24V:9 | 24V | 9 | -B22:97 | |
| -X2:24V:10 | 24V | 10 | -Q21:13/NO | |
| -X2:GND:1 | GND | 1 | -T1:- | |
| -X2:GND:2 | GND | 2 | -C1:0V | |
| -X2:GND:3 | GND | 3 | -C1:- | |
| -X2:GND:4 | GND | 4 | -Q11:A2 | |
| -X2:GND:5 | GND | 5 | -P1:X2 | |
| -X2:GND:6 | GND | 6 | -Q21:A2 | |
| -X2:GND:7 | GND | 7 | -P2:X2 | |

## -X3

| Designation | Group | Index | Side A | Side B |
|---|---|---|---|---|
| -X3:PE:1 | PE | 1 | -X1:PE:1 | +EXT-M1:PE |
| -X3:PE:2 | PE | 2 | -X1:PE:1 | +EXT-M2:PE |
| -X3:U:1 | U | 1 | -B12:2/T1 | +EXT-M1:U1 |
| -X3:U:2 | U | 2 | -B22:2/T1 | +EXT-M2:U1 |
| -X3:V:1 | V | 1 | -B12:4/T2 | +EXT-M1:V1 |
| -X3:V:2 | V | 2 | -B22:4/T2 | +EXT-M2:V1 |
| -X3:W:1 | W | 1 | -B12:6/T3 | +EXT-M1:W1 |
| -X3:W:2 | W | 2 | -B22:6/T3 | +EXT-M2:W1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.6 | Revision date: 2026-10-06 | |
| Customer | Author: fransys-examples | Page title: Terminal list | Scope: Pump cabinet | Sheet | Page: 7 of 8 | |

| Mpn | Revision | Manufacturer | Description | Count | Designations |
|---|---|---|---|---|---|
| 1757022 | | Phoenix Contact | PCB connector plug, 3-pin, 5.08 mm pitch | 2 | -J1; -J2 |
| 2002-1201 | | WAGO | TOPJOB S feed-through terminal block, 2.5 mm2, grey; 24 A; 800 V | 39 | -X01:L1:1; -X01:L1:2; -X01:L1:3; -X01:L1:4; -X01:L2:1; -X01:L2:2; -X01:L2:3; -X01:L2:4; -X01:L3:1; -X01:L3:2; -X01:L3:3; -X01:L3:4; -X1:L1:1; -X1:L2:1; -X1:L3:1; -X1:N:1; -X2:24V:1; -X2:24V:2; -X2:24V:3; -X2:24V:4; -X2:24V:5; -X2:24V:6; -X2:24V:7; -X2:24V:8; -X2:24V:9; -X2:24V:10; -X2:GND:1; -X2:GND:2; -X2:GND:3; -X2:GND:4; -X2:GND:5; -X2:GND:6; -X2:GND:7; -X3:U:1; -X3:U:2; -X3:V:1; -X3:V:2; -X3:W:1; -X3:W:2 |
| 2002-1207 | | WAGO | TOPJOB S ground terminal block, 2.5 mm2, green-yellow | 3 | -X1:PE:1; -X3:PE:1; -X3:PE:2 |
| 3LD2054-0TK51 | | Siemens | Switch-disconnector, 3-pole; 16 A | 1 | -Q1 |
| 3NW7033 | | Siemens | Fuse holder, 3-pole, 10x38 mm; 32 A | 2 | -F11; -F21 |
| 3NW8004-1 | | Siemens | SENTRON cylindrical fuse link, 10x38 mm, aM; 4 A | 6 | -F11; -F21 |
| 3SU1102-6AA40-1AA0 | | Siemens | Pilot light 22 mm, green, integrated LED 24 V AC/DC | 2 | -P1; -P2 |
| 5SY6506-7 | | Siemens | Miniature circuit breaker, 1P+N, curve C; 6 A | 1 | -F01 |
| 750-402 | | WAGO | 4-channel digital input module, 24 V DC | 1 | -DI1 |
| 750-504 | | WAGO | 4-channel digital output module, 24 V DC; 0.5 A | 1 | -DO1 |
| 750-600 | | WAGO | End module, terminates the internal data bus; no data contacts | 1 | -E1 |
| 750-8212 | | WAGO | PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485 | 1 | -C1 |
| LADN11 | | Schneider Electric | TeSys D front-mount auxiliary contact block, 1 NO + 1 NC | 2 | -Q11; -Q21 |
| LC1D09BD | | Schneider Electric | TeSys D contactor, 3-pole, AC-3 9 A, 1 NO + 1 NC auxiliary, 24 V DC coil | 2 | -Q11; -Q21 |
| LRD08 | | Schneider Electric | TeSys thermal overload relay, class 10A, 2.5-4 A, 1 NO + 1 NC | 2 | -B12; -B22 |
| QUINT4-PS/1AC/24DC/2.5/SC | | Phoenix Contact | Power supply, 1-phase, 24 V DC; 2.5 A | 1 | -T1 |
| SKX-RIB-2 | 1.3 | | Relay interface board | 1 | -U2 |

| | | | | | |
|---|---|---|---|---|---|
| Title<br>**Pump cabinet** | | | Number<br>SKX-PC-2 | Revision<br>1.6 | Revision date<br>2026-10-06 |
| Customer | Author<br>fransys-examples | Page title<br>BOM | Scope<br>Pump cabinet | Sheet | Page<br>8 of 8 |